25901264 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFSQ200 - FOOD PACKAGING TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Trays made from thermoformed plastic are termed? [U] / [CO1]

   a). Semi-rigid packaging b). Flexible packaging

   c). Rigid packaging d). Foil packaging

2. A screw closure is mainly used for which purpose? [U] / [CO2]

   a). Decorate b). Sealing containers

   c). Labeling d). Cushioning

3. Dry lamination is preferred for food packaging because : [U] / [CO2]

   a). It provide strong and stable bonds b). It is cheap

   c). It use water d). It requires no heat

4. Select the difference between global migration and specific migration. [K] / [CO4]

a). Specific migration measures individual chemicals
b). Both measure same parameters

c). Global measures individual substances
d). Specific migration is not regulated

5. Hot melt lamination is often preferred over solvent lamination because it is_____. [U] / [CO2]

a). Use harmful solvents
b). Is faster and solvent free

c). Can't bond paper
d). Needs low temperature

6. Interpret which packaging mainly provides the first level of protection to food. [U] / [CO1]

a). Secondary
b). Primary

c). Teritiary
d). Quarternary

7. Main disadvantage of paper packaging is: [U] / [CO2]

a). Moisture sensitivity
b). Odour

c). Weight
d). Toxicity

8. RFID stand for_____-. [U] / [CO3]

a). Radio Frequency Identification
b). Radio Function Development

c). Radio Field identity
d). None of these

9. Describe the principle of Gas Transmission Rate testing. [U] / [CO4]

a). Measuring gas flow through a material under controlled conditions
b). Measuring tensile strength of packaging

c). Measuring moisture absorption
d). Measuring package weight

10. Select the scope of food packaging includes supporting export of food by___. [K] / [CO1]

a). Increasing food moisture
b). Changing food formulation

c). Meeting international packaging standards
d). Reducing food weight

11. Polypropylene is widely used in food packaging because of its: [U] / [CO2]

a). High melting point and chemical resistance
b). High density

c). Poor heat resistance
d). Brittleness

12. The ability of plastic resist deformation under load is known as ? [U] / [CO2]

a). Flexibility b). Toughness

c). Rigidity d). Transparency

13. The sealing method used for retort pouches. [U] / [CO3]

a). Adhesive b). Stitching

c). Heat sealing d). Clip sealing

14. Select the packaging material property mainly assessed by puncture resistance. [U] / [CO4]

a). Barrier property b). Mechanical integrity

c). Optical Clarity d). Thermal resistance

15. Define the purpose of protective lacquers inside food cans. [K] / [CO2]

a). To improve can colour b). To prevent metal–food reaction

c). To reduce can weight d). To increase transparency

16. Identify the factor that does not affect WVTR. [K] / [CO4]

a). Material thickness b). Relative humidity

c). Temperature d). Package color

17. Heat sealing property in laminates is mainly provided by: [U] / [CO2]

a). Metal layer b). Plastic layer

c). Paper layer d). Adhesive layer

18. Oxygen is intentionally included in MAP for packaging of_____. [U] / [CO3]

a). Cheese b). Bakery products

c). Fresh red meat d). Dry fruits

19. Global migration in food packaging refers to [K] / [CO4]

a). Movement of gases through packaging b). Total transfer of substances from packaging into food

c). Loss of nutrients from food d). Permeability of packaging material

20. List which role of packaging helps consumers make informed food choices. [K] / [CO1]

a). Labelling information b). Heating

c). Decoration d). Food colouring

[1x20 = 20]

Part B

Short Answer

Answer any **6** questions

Each question carries **5** marks

21. Explain the legal control of misleading claims on food packages. [A] / [CO3]

22. Outline why amber and green glass are preferred for packaging light-sensitive food and beverage products. [K] / [CO2]

23. Write a short note on biodegradable plastics. [U] / [CO2]

24. Discuss the role of tensile strength in handling and transportation of packaged foods. [U] / [CO4]

25. Describe tin-free steel (TFS) and mention its advantages over tinplate. [U] / [CO2]

26. Distinguish vibration and compression test used during transportation. [An] / [CO4]

27. Describe the pouches used in food packaging. [U] / [CO2]

28. Write a note on types of form - fill- seal packaging machines. [U] / [CO3]

[5x6 = 30]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

29. Explain bursting strength of packaging materials. Discuss on factors affecting and its importance in food packaging applications. [U] / [CO4]

30. Describe the manufacture of paper for packaging purpose and discuss how processing affects its properties. [U] / [CO2]

31. Outline the protective, containment, and communication functions of food packaging in the food industry [K] / [CO1]

32. Describe paper types,uses and its applications. [U] / [CO2]

[10x2 = 20]